PADBen: A Comprehensive Benchmark for Evaluating AI Text Detectors Against Paraphrase Attacks

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Abstract. While AI-generated text (AIGT) detectors achieve over 90% accuracy on direct LLM outputs, they fail against iteratively-paraphrased content. This paper investigates why detectors fail through intrinsic mechanism exploration, establishing two key findings: (1) LLM-paraphrased text constitutes a qualitatively distinct third category in representation space, separate from both human-authored and LLM-generated text; (2) iterative paraphrasing induces cumulative semantic drift away from the source at a decelerating rate, while generation-level patterns in hidden states remain stable. These findings explain why paraphrase attacks succeed: text is relocated into regions that binary AIGT classifiers have not been trained to handle, while retaining origin-dependent generation signatures underlying two distinct attack categories—*authorship obfuscation* (paraphrasing human-authored text) and *plagiarism evasion* (paraphrasing LLM-generated text).

To address these vulnerabilities, PADBen (**P**araphrase **A**ttack **D**etection **Ben**chmark) is introduced—the first benchmark systematically evaluating detector robustness against both attack scenarios. PADBen comprises a five-type text taxonomy and five progressive detection tasks across sentence-pair and single-sentence formats. Evaluation of 11 state-of-the-art detectors reveals critical asymmetry: detectors can identify plagiarism evasion but largely fail on authorship obfuscation, underscoring the need for detection methods beyond binary human-vs-AIGT discrimination.

Keywords: New Topics in Artificial Intelligence and Machine Learning · Explainable AI · Large Language Models

1 Introduction

Large Language Models (LLMs) have achieved near-human quality in text generation [1,2,3], enabling unprecedented automation while posing significant risks through misinformation and spam [4,5]. This has spurred development of AI-generated text (AIGT) detectors, spanning zero-shot methods (FastDetect-GPT [6], DetectGPT [7], GLTR [8], Binoculars [9]) and model-based detectors (RADAR [10], OpenAI RoBERTa [11]).

Paraphrase attacks have emerged as the most effective evasion strategy [12,13,14,15], systematically rewording AI-generated content while preserving semantic meaning to evade detection [16,17], causing detector accuracy to plummet to near-random performance [18,19].

Yet existing benchmarks treat paraphrasing as one perturbation among many. RAID [20] employs only single-step Dipper-based paraphrasing; MAGE [21] omits adversarial evasion entirely; complementary works on multilingual detection [22], QA [23] and scientific text [24] shares the same gap. PARAPHRASUS [25] focuses on paraphrase *identification* (whether there is a paraphrased text) rather than detector robustness against iterative evasion(whether the detectors can detect paraphrase attack). Crucially, no existing framework distinguishes authorship obfuscation from plagiarism evasion, nor explains the intrinsic mechanism behind paraphrase attack success.

To address this gap, PADBen (**P**araphrase **A**ttack **D**etection **Ben**chmark) is introduced. Through intrinsic mechanism analysis (Section 2), we establish: **(1) Semantic Displacement**—LLM-paraphrased text constitutes a qualitatively distinct third category in representation space; **(2) Decelerating Drift with Stable Generation Patterns**—iterative paraphrasing induces cumulative semantic drift away from the source while generation-level patterns in hidden states remain stable, with the steepest trajectory within the first 3 iterations. Together, these explain why paraphrase attacks succeed: text is relocated into regions binary classifiers cannot handle, while retaining origin-dependent signatures that produce asymmetric detection outcomes. Built on a five-type text taxonomy and five progressive detection tasks, PADBen evaluates 11 detectors across sentence-pair and single-sentence formats. Code, datasets, and supplementary appendices are available at `https://github.com/JonathanZha47/PadBen-Paraphrase-Attack-Benchmark`.
The key contributions are:

1. **Intrinsic Mechanism Analysis.** First formal investigation of paraphrase attack mechanisms, establishing semantic displacement and decelerating drift with stable generation patterns—enabling two distinct attack categories: *authorship obfuscation* and *plagiarism evasion*;
2. **Benchmark Construction.** A five-type taxonomy and five progressive tasks evaluating detector robustness across both attack scenarios;
3. **Comprehensive Evaluation.** Evaluation of 11 detectors (4 zero-shot, 7 model-based) across five tasks reveals critical asymmetry: plagiarism evasion remains detectable (RADAR AUC 0.909) while authorship obfuscation substantially degrades all detectors (best AUC 0.748, near-random for most), demonstrating that current binary human-vs-AIGT classifiers are fundamentally ill-suited for the paraphrase attack setting.

2 How Do Paraphrase Attacks Intrinsically Work?

Since iteratively-paraphrased text is also AI-generated, why do paraphrase attacks evade AIGT detection systems?
Current AIGT detectors are trained to distinguish human-authored text from LLM-generated text—but paraphrasing is neither: it is a transformation applied

on top of existing text regardless of origin. We claim that this manifests itself as a qualitatively different trajectory in the representation space, causing detectors to fail.

2.1 Hypothesis 1 Discussion

To validate this claim, we initially investigate whether paraphrasing text is different from existing text types, and hence create unique attack vectors that existing detectors hard to capture.

Notation. Let $x_{\text{human}} \in \mathcal{X}_{\text{human}}$, $x_{\text{gen}} \in \mathcal{X}_{\text{gen}}$, and $x_{\text{para}} \in \mathcal{X}_{\text{para}}$ denote text instances from the sets of human-authored, LLM-generated, and LLM-paraphrased texts, respectively. Let $\phi : \mathcal{X} \to \mathbb{R}^d$ be a representation function that maps text to a d-dimensional vector space, and let $\text{dist}(\cdot, \cdot)$ denote a semantic similarity distance between two representations. The population centroid of a text set $\mathcal{X}_t$ under ϕ is:

$$\bar{\phi}(\mathcal{X}_t) = \frac{1}{|\mathcal{X}_t|} \sum_{x \in \mathcal{X}_t} \phi(x) \tag{1}$$

Hypothesis 1 *LLM-paraphrased text occupies a distinct region in ϕ-space, separate from both human-authored and LLM-generated text. Specifically, the three text categories form a non-degenerate configuration in representation space:*

$$dist\big(\bar{\phi}(\mathcal{X}_{para}), \bar{\phi}(\mathcal{X}_{human})\big) \neq dist\big(\bar{\phi}(\mathcal{X}_{para}), \bar{\phi}(\mathcal{X}_{gen})\big) \neq dist\big(\bar{\phi}(\mathcal{X}_{human}), \bar{\phi}(\mathcal{X}_{gen})\big) \tag{2}$$

If this holds, $\mathcal{X}_{\text{para}}$ is not simply a noisy variant of $\mathcal{X}_{\text{gen}}$—it occupies an intermediate semantic region that binary human-vs-AIGT classifiers are not trained to handle, explaining why detectors trained on $(\mathcal{X}_{\text{human}}, \mathcal{X}_{\text{gen}})$ pairs fail under paraphrase attacks.

Experiment 1 Setup.

Three text categories are constructed from 16,233 human-authored sentences sourced from three established paraphrase datasets (details in Appendix A):

(1) $\mathcal{X}_{\text{human}}$ — the original human-authored sentences;

(2) $\mathcal{X}_{\text{gen}}$ — LLM-generated texts produced via semantic equivalence prompting using GPT-4o, instructed to preserve meaning and structure *without* explicit rewording;

(3) $\mathcal{X}_{\text{para}}$ — LLM-paraphrased texts generated from the same source sentences using explicit paraphrasing instructions via GPT-4o.

The key distinction between (2) and (3) lies in the intent of generation : $\mathcal{X}_{\text{gen}}$ focuses on the "same semantic meaning," while $\mathcal{X}_{\text{para}}$ explicitly focuses on "paraphrasing" it. BGE-M3 embeddings are computed for all three categories, and pairwise distances $\text{dist}(\bar{\phi}(\mathcal{X}_i), \bar{\phi}(\mathcal{X}_j))$ are measured in a full-dimensional space. The PCA projection in 2D and K-means clustering ($k = 3$) is additionally applied to assess geometric separability. Full detailed setups are provided in Appendix B.

Experiment 1 Results.

Table 1: Pairwise cosine similarity between text categories in BGE-M3 embedding space. Values are averaged over 5 independent runs.

Comparison	Cosine
$\mathcal{X}_{\text{human}} \leftrightarrow \mathcal{X}_{\text{gen}}$	0.195
$\mathcal{X}_{\text{human}} \leftrightarrow \mathcal{X}_{\text{para}}$	**0.068**
$\mathcal{X}_{\text{gen}} \leftrightarrow \mathcal{X}_{\text{para}}$	0.214

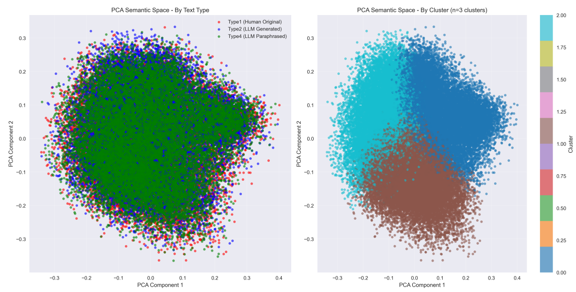

Fig. 1: PCA projection of semantic space (left) and K-means clustering results (right, $k = 3$). Despite measurable distance differences (Table 1), text categories show substantial overlap in 2D projection.

To reduce the effect of sampling variance, all distance measurements are averaged over 5 independent runs. Table 1 (full table is available in Appendix B) reports the mean pairwise distances between the three centroids of the text category. The empirical results yield the following ordering:

$$\text{dist}\big(\bar{\phi}(\mathcal{X}_{\text{para}}), \bar{\phi}(\mathcal{X}_{\text{human}})\big) \;<\; \text{dist}\big(\bar{\phi}(\mathcal{X}_{\text{human}}), \bar{\phi}(\mathcal{X}_{\text{gen}})\big) \;<\; \text{dist}\big(\bar{\phi}(\mathcal{X}_{\text{para}}), \bar{\phi}(\mathcal{X}_{\text{gen}})\big) \tag{3}$$

i.e., $0.068 < 0.195 < 0.214$, which confirms Hypothesis 1: $\mathcal{X}_{\text{para}}$ constitutes a qualitatively distinct text category.

An interesting finding emerges from the PCA projection of the representation space (Figure 1): despite the clear distance ordering confirmed in Equation 3, all three categories show substantial overlap in 2D PCA space, with K-means clustering failing to recover text-type-aligned boundaries. This indicates that the discriminative signal between $\mathcal{X}_{\text{human}}$, $\mathcal{X}_{\text{gen}}$, and $\mathcal{X}_{\text{para}}$ is distributed across many high-dimensional directions simultaneously, rather than concentrating along the first two principal components. This explains why zero-shot statistical detectors—which operate on low-dimensional statistical summaries of text—fail to capture the separation between these categories, as we confirm empirically in Section 5.

2.2 Hypothesis 2 Discussion

Hypothesis 1 establishes that $\mathcal{X}_{\text{para}}$ occupies a distinct region in ϕ-space that lies closer to $\mathcal{X}_{\text{human}}$ than to $\mathcal{X}_{\text{gen}}$. This raises a natural follow-up question: if a single paraphrasing step already moves text toward the human-authored region, does *iterative* paraphrasing continue this trajectory, progressively making text more human-like and thus increasingly harder to detect?

Notation. Let $x^{(0)} \in \mathcal{X}$ denote an original text, and let $x^{(k)} = \text{Para}^{(k)}(x^{(0)})$ denote the text after k iterations of LLM paraphrasing, where each iteration applies the paraphrasing operation to the output of the previous one. For a text

Table 2: Cosine similarity between iteratively paraphrased text and reference centroids $\bar{\phi}^{(0)}(\mathcal{X}_{\text{human}})$ and $\bar{\phi}^{(0)}(\mathcal{X}_{\text{gen}})$ in BGE-M3 embedding space. Full table in Appendix B.

Ref.	Origin	Iteration k				
		2	4	6	8	10
$\bar{\phi}^{(0)}_{\text{human}}$	$\mathcal{X}_{\text{human}}$	0.085	0.107	0.122	0.128	0.134
	$\mathcal{X}_{\text{gen}}$	0.698	0.697	0.697	0.699	0.698

Table 3: Cosine similarity between consecutive iterations $\text{dist}(\bar{\phi}^{(k)}(\mathcal{X}_t), \bar{\phi}^{(k+1)}(\mathcal{X}_t))$ for $t \in \{\mathcal{X}_{\text{human}}, \mathcal{X}_{\text{gen}}\}$ under BGE-M3 embeddings and Qwen3-4B hidden states.

Repr.	Origin	Iteration k				
		2	4	6	8	10
BGE Emb.	$\mathcal{X}_{\text{human}}$	0.048	0.072	0.083	0.092	0.100
	$\mathcal{X}_{\text{gen}}$	0.047	0.068	0.077	0.087	0.091
Hid. Stat.	$\mathcal{X}_{\text{human}}$	0.012	0.022	0.015	0.017	0.019
	$\mathcal{X}_{\text{gen}}$	0.011	0.014	0.015	0.017	0.018

set $\mathcal{X}_t$ of origin type $t \in \{\text{human}, \text{gen}\}$, the centroid after k iterations is:

$$\bar{\phi}^{(k)}(\mathcal{X}_t) = \frac{1}{|\mathcal{X}_t|} \sum_{x \in \mathcal{X}_t} \phi\Big(\text{Para}^{(k)}(x)\Big) \tag{4}$$

where $\bar{\phi}^{(0)}(\mathcal{X}_t) = \bar{\phi}(\mathcal{X}_t)$ denotes the centroid of the original unparaphrased texts.

Hypothesis 2 *Iterative paraphrasing progressively moves text toward $\mathcal{X}_{human}$, making it increasingly indistinguishable from human-authored content. Specifically, the distance between iteratively paraphrased text and $\mathcal{X}_{human}$ decreases monotonically with iteration count k:*

$$dist\Big(\bar{\phi}^{(k+1)}(\mathcal{X}_t),\, \bar{\phi}^{(0)}(\mathcal{X}_{human})\Big) \;<\; dist\Big(\bar{\phi}^{(k)}(\mathcal{X}_t),\, \bar{\phi}^{(0)}(\mathcal{X}_{human})\Big) \tag{5}$$

for both $t \in \{human, gen\}$ and all $k \geq 0$.

Experiment 2 Setup. 1000 texts are sampled from each of $\mathcal{X}_{\text{human}}$ and $\mathcal{X}_{\text{gen}}$, with $k = 10$ paraphrasing iterations performed using Qwen3-4B-Instruct, yielding iteratively paraphrased sets $\{x^{(k)}\}_{k=1}^{10}$ for each origin type. For each iteration, the following are extracted: (1) paraphrased text, (2) final layer hidden states (4096-dim) with mean pooling, and (3) BGE-M3 embeddings (1024-dim). Cosine similarities are computed between iteration centroids and reference centroids $\bar{\phi}^{(0)}(\mathcal{X}_{\text{human}})$ and $\bar{\phi}^{(0)}(\mathcal{X}_{\text{gen}})$, with PCA applied to centroid trajectories (Appendix B).

Experiment 2 Results.
Table 2 directly tests Hypothesis 2. For $\mathcal{X}_{\text{human}}$, the cosine similarity to $\bar{\phi}^{(0)}(\mathcal{X}_{\text{human}})$ increases monotonically with k ($0.085 \rightarrow 0.134$), and for $\mathcal{X}_{\text{gen}}$, the cosine similarity to $\bar{\phi}^{(0)}(\mathcal{X}_{\text{human}})$ remains stable (≈ 0.698 across all iterations). Both observations **reject Hypothesis 2**—iterative paraphrasing does not move text progressively closer to $\mathcal{X}_{\text{human}}$; instead it drifts *away* from its own origin while maintaining a constant distance from $\mathcal{X}_{\text{gen}}$.
Table 3 further examines the inter-iteration drift dynamics $\text{dist}(\bar{\phi}^{(k)}(\mathcal{X}_t), \bar{\phi}^{(k+1)}(\mathcal{X}_t))$ for $t \in \{\mathcal{X}_{\text{human}}, \mathcal{X}_{\text{gen}}\}$, $k \geq 1$, revealing two key patterns.

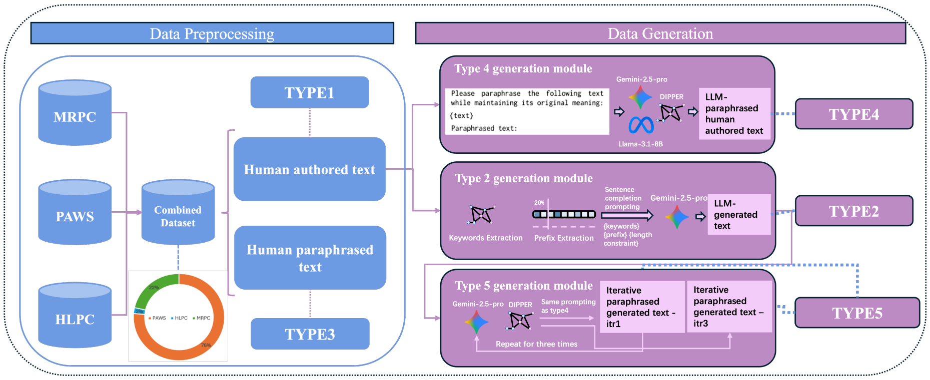

Fig. 2: Overall pipeline for benchmark curation. Preprocessing details in Appendix A, data generation in Appendix C.

(1) Decelerating Semantic Drift. Under BGE-M3 embeddings, the inter-iteration cosine similarity accumulates for both $\mathcal{X}_{\text{human}}$ ($0.048 \to 0.100$) and $\mathcal{X}_{\text{gen}}$ ($0.047 \to 0.091$), confirming cumulative semantic displacement with each paraphrasing step. Additionally, the *rate* of displacement decelerates as k increases—the gain from iteration 1 to 4 (0.072) is substantially larger than from iteration 4 to 10 (0.028 for $\mathcal{X}_{\text{human}}$). This diminishing return indicates that beyond a certain depth, additional paraphrasing iterations introduce only trivial semantic change while risking incoherence in the generated text. This motivates our benchmark design choice to cap at 1-iteration and 3-iteration variants in PADBen.

(2) Representation-Dependent Drift Magnitude. BGE-M3 embeddings exhibit substantially larger inter-iteration displacement (0.047–0.100) than Qwen3-4B hidden states (0.011–0.022). This divergence reflects fundamental differences in what each representation captures—BGE-M3's contrastive training is optimized to track *semantic content* variations, while Qwen3-4B hidden states encode *generation-level patterns* (lexical choices, syntactic constructions, stylistic regularities) learned during causal language model pre-training. The stability of hidden-state distances across iterations therefore provides evidence that generation-level patterns are more resistant to paraphrase perturbation than semantic content.

Together, these two patterns lead to a revised understanding: rather than moving text toward $\mathcal{X}_{\text{human}}$ as Hypothesis 2 predicted, iterative paraphrasing induces *cumulative semantic drift away from the source* while *preserving generation-level patterns*. This mechanism enables two distinct attack scenarios. **(1) Authorship Obfuscation:** Human-authored text undergoing iterative paraphrasing maintains human-like stylistic markers despite semantic drift, creating detection blind spots that enable unauthorized appropriation of human writing. **(2) Plagiarism Evasion:** LLM-generated text experiencing iterative paraphrasing preserves AI-like generation patterns while achieving sufficient semantic transformation to evade plagiarism detection systems, facilitating academic misconduct.

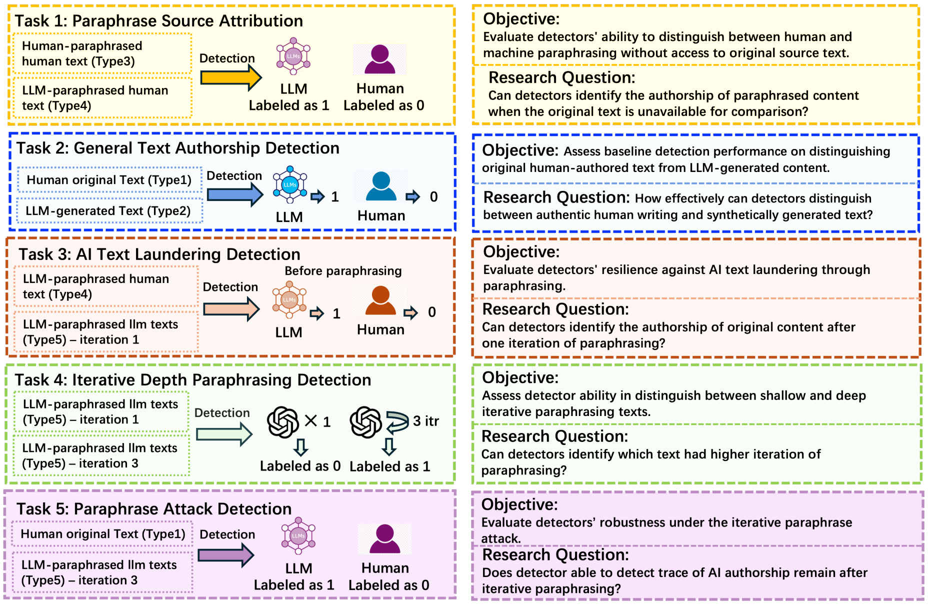

Fig. 3: Overall Task introduction for the Benchmark. Task 1-5 measures the detector's different capabilities covering the robustness, performance when encountering the paraphrase attack. Detailed task specifics can be found in Appendix C.

3 Benchmark Curation

Text Type Taxonomy. The two attack scenarios identified in Section 2 map onto distinct real-world threats: (a) *Authorship obfuscation*: a writer paraphrases human-authored text through an LLM to evade stylometric attribution or plagiarism detection—the paraphrase must appear human-authored. (b) *Plagiarism evasion*: an author paraphrases LLM-generated content iteratively to remove detectable AI artifacts before submission—the output must evade AIGT classifiers. These two scenarios require fundamentally different text collections to evaluate: the former compares human paraphrases (Type 3) vs. LLM paraphrases (Type 4) of human text; the latter contrasts deeply laundered AI text (Type 5) against baselines (Types 1, 2). The taxonomy is therefore grounded in these two threat scenarios and is not ad-hoc. A five-category taxonomy is established: **Type 1** — Human original text; **Type 2** — LLM-generated text; **Type 3** — Human-paraphrased human text; **Type 4** — LLM-paraphrased human text; **Type 5** — LLM-iteratively-paraphrased LLM text. Detailed definitions can be found in Appendix C.

Data Preparation. The benchmark leverages three established datasets: Microsoft Research Paraphrase Corpus (**MRPC**) [26], Human-LLM Paraphrase Corpus (**HLPC**) [27], and Paraphrase Adversaries from Word Scrambling (**PAWS**) [28]. A cosine similarity filter (threshold: 0.85) is applied to remove near-duplicates,

Table 4: Zero-shot Detectors Performance Summary

MODEL	CHALLENGE	METHOD	Task 1				Task 2				Task 3				Task 4				Task 5			
			AUC	T1	T5	T10	AUC	T1	T5	T10	AUC	T1	T5	T10	AUC	T1	T5	T10	AUC	T1	T5	T10
BINOCULAR	sentence-pair		0.399	0.003	0.023	0.050	0.457	0.007	0.037	0.081	0.505	0.008	0.046	0.099	0.498	0.011	0.052	0.106	0.457	0.006	0.032	0.070
	single-sentence	exhaustive	0.439	0.011	0.046	0.080	0.579	0.029	0.106	0.184	0.500	0.008	0.050	0.101	0.486	0.012	0.050	0.098	0.428	0.009	0.047	0.087
	single-sentence	sampling_30%	0.437	0.010	0.046	0.081	0.575	0.031	0.109	0.188	0.495	0.008	0.050	0.096	0.486	0.010	0.051	0.098	0.429	0.008	0.050	0.090
	single-sentence	sampling_50%	0.443	0.011	0.047	0.084	0.583	0.032	0.112	0.194	0.499	0.008	0.051	0.102	0.487	0.012	0.053	0.099	0.434	0.009	0.051	0.092
	single-sentence	sampling_80%	0.453	0.010	0.049	0.087	0.589	0.034	0.116	0.195	0.507	0.007	0.058	0.106	0.487	**0.016**	**0.060**	0.104	0.439	0.012	0.053	0.092
FAST_DETECT_GPT	sentence-pair		0.638	**0.027**	**0.115**	0.204	0.787	0.086	0.249	0.369	0.503	0.012	0.055	0.108	0.476	0.005	0.036	0.081	0.606	0.023	0.090	0.165
	single-sentence	exhaustive	0.573	0.006	0.044	0.099	0.665	0.010	0.075	0.149	0.504	0.011	0.051	0.103	0.488	0.009	0.051	0.099	0.568	0.006	0.047	0.103
	single-sentence	sampling_30%	0.576	0.006	0.046	0.097	0.666	0.009	0.078	0.154	0.502	0.011	0.049	0.096	0.490	0.007	0.046	0.091	0.572	0.005	0.045	0.100
	single-sentence	sampling_50%	0.581	0.005	0.045	0.098	0.672	0.010	0.078	0.153	0.505	0.010	0.051	0.101	0.491	0.008	0.049	0.095	0.577	0.005	0.047	0.102
	single-sentence	sampling_80%	0.587	0.004	0.040	0.092	0.675	0.008	0.076	0.151	0.514	0.007	0.045	0.100	0.488	0.011	0.049	0.095	0.578	0.005	0.048	0.103
GLTR	sentence-pair		0.429	0.007	0.043	0.082	0.436	0.006	0.045	0.086	0.529	0.011	0.060	0.116	0.514	**0.016**	**0.057**	**0.113**	0.482	0.012	0.054	0.108
	single-sentence	exhaustive	0.459	0.006	0.032	0.068	0.480	0.004	0.021	0.056	0.513	0.012	0.059	0.122	0.506	**0.013**	**0.057**	**0.113**	0.488	0.011	0.047	0.091
	single-sentence	sampling_30%	0.457	0.004	0.034	0.066	0.474	0.003	0.019	0.053	0.519	0.012	0.065	0.124	0.502	**0.014**	**0.056**	**0.109**	0.484	0.012	0.045	0.085
	single-sentence	sampling_50%	0.461	0.005	0.036	0.068	0.480	0.004	0.022	0.057	0.524	0.012	0.066	0.126	**0.507**	**0.015**	**0.059**	**0.114**	0.489	0.011	0.049	0.089
	single-sentence	sampling_80%	0.458	0.006	0.031	0.063	0.482	0.004	0.021	0.059	0.523	0.011	0.062	0.117	**0.509**	0.013	0.059	**0.111**	0.491	0.011	0.047	0.095
RADAR	sentence-pair		**0.728**	0.004	0.105	**0.246**	**0.910**	**0.142**	**0.566**	**0.809**	**0.748**	**0.054**	**0.234**	**0.372**	**0.526**	0.010	0.055	0.112	**0.909**	**0.140**	**0.542**	**0.808**
	single-sentence	exhaustive	**0.648**	**0.038**	**0.190**	**0.345**	**0.793**	**0.063**	**0.313**	**0.567**	**0.633**	**0.016**	**0.080**	**0.160**	**0.511**	0.010	0.052	0.104	**0.797**	**0.062**	**0.310**	**0.562**
	single-sentence	sampling_30%	**0.642**	**0.037**	**0.187**	**0.337**	**0.789**	**0.063**	**0.313**	**0.560**	**0.627**	**0.016**	**0.078**	**0.157**	**0.508**	0.010	0.051	0.103	**0.797**	**0.062**	**0.312**	**0.560**
	single-sentence	sampling_50%	**0.644**	**0.036**	**0.181**	**0.337**	**0.789**	**0.060**	**0.302**	**0.559**	**0.628**	**0.016**	**0.078**	**0.155**	0.506	0.010	0.051	0.102	**0.795**	**0.060**	**0.300**	**0.556**
	single-sentence	sampling_80%	**0.648**	**0.039**	**0.195**	**0.345**	**0.797**	**0.067**	**0.335**	**0.569**	**0.630**	**0.016**	**0.078**	**0.156**	0.508	0.010	0.051	0.102	**0.803**	**0.066**	**0.332**	**0.568**

Note: AUC = AUC-ROC, T1 = TPR@1%FPR, T5 = TPR@5%FPR, T10 = TPR@10%FPR. Best (**red bold**) and second-best (blue underlined) results are marked within each setup for each task and metric.

combining them to obtain 16,233 human-authored texts (Type 1) and human-paraphrased human texts (Type 3). Figure 2 illustrates the detailed procedure of how raw data is preprocessed and how Type 2, 4, and 5 texts are generated via a modularized generation pipeline: **Type 2** via sentence completion using Google Gemini-2.5-Pro; **Type 4** via multi-model paraphrasing (DIPPER, Gemini-2.5-Pro, LLaMA-3-8B); and **Type 5** via iterative paraphrasing with temperature scaling and convergence detection.

Quality Assurance. Dataset quality is verified using three metrics: Jaccard similarity (inter-type semantic preservation), perplexity (linguistic diversity), and self-BLEU (intra-type diversity). Full analysis, including comparison with the RAID benchmark, is provided in Appendix C. PADBen achieves substantially higher diversity and perplexity than RAID.

Task Introduction. Five progressive detection tasks are designed to systematically assess detector's robustness in varying complexity levels and attack scenarios. Each task targets specific vulnerabilities while leveraging the multi-type text dataset to evaluate detectors under increasingly sophisticated adversarial conditions. Figure 3 demonstrates the description of the five tasks, and Appendix C provides further details on task objectives.

4 Evaluation Framework

Two categories of detectors are examined: **1. Zero-shot detectors** Operating without task-specific training, relying on pre-existing linguistic/statistical patterns to distinguish human vs. machine text; the evaluated detectors include Binoculars, Fast-Detect-GPT, GLTR, and RADAR (Appendix E). **2. Model-based detectors** Leveraging pre-trained language models or instruction-based LLMs for classification, using few-shot and persona prompting strategies for both evaluation challenges (Appendix E).

Evaluation Setup. Evaluation is conducted under two challenges—single-sentence classification and sentence-pair recognition—across 5 setups to reflect

Table 5: Model-based Detectors Performance Summary

Model	Challenge	Task 1				Task 2				Task 3				Task 4				Task 5			
		AUC	T1	T5	T10	AUC	T1	T5	T10	AUC	T1	T5	T10	AUC	T1	T5	T10	AUC	T1	T5	T10
Claude-3.5-Haiku	sentence-pair	0.623	0.016	0.078	0.155	0.366	0.005	0.024	0.047	0.475	0.010	0.042	0.085	0.507	0.010	0.051	0.102	0.351	0.003	0.017	0.034
	single-sentence	**0.514**	**0.012**	0.058	**0.115**	0.535	**0.018**	**0.089**	**0.162**	0.519	0.022	0.069	0.069	0.503	0.011	0.054	0.073	0.545	**0.049**	0.112	0.112
DeepSeek-V2.5	sentence-pair	0.572	0.012	0.060	0.120	0.467	0.008	0.039	0.078	**0.519**	**0.011**	**0.057**	**0.113**	**0.514**	**0.011**	**0.054**	**0.108**	0.484	0.009	0.046	0.091
	single-sentence	0.480	0.009	0.046	0.092	0.479	0.009	0.047	0.095	0.531	0.014	0.069	**0.138**	0.501	0.010	0.051	0.101	0.511	0.011	0.055	0.110
Gemma-3-27B	sentence-pair	0.521	0.011	0.053	0.105	0.506	0.010	0.051	0.102	0.480	0.010	0.048	0.095	0.502	0.010	0.050	0.101	0.516	0.010	0.052	0.104
	single-sentence	0.461	0.008	0.040	0.080	0.465	0.009	0.044	0.088	0.520	0.013	0.064	0.129	0.503	0.010	0.052	0.104	0.478	0.009	0.043	0.085
Kimi-K2-Instruct	sentence-pair	**0.691**	**0.020**	**0.102**	**0.204**	0.431	0.007	0.036	0.071	0.516	**0.011**	0.053	0.106	0.487	0.010	0.048	0.096	0.441	0.006	0.030	0.060
	single-sentence	0.509	**0.012**	**0.062**	0.096	**0.540**	0.014	0.071	0.141	**0.540**	**0.029**	**0.123**	0.123	0.503	0.011	**0.056**	0.063	**0.573**	0.047	**0.185**	**0.185**
Llama-4-Scout -17B	sentence-pair	0.561	0.012	0.059	0.118	0.456	0.008	0.042	0.085	**0.519**	**0.011**	0.054	0.109	0.493	0.010	0.049	0.098	0.476	0.009	0.046	0.091
	single-sentence	0.486	0.009	0.047	0.095	0.472	0.009	0.046	0.091	0.523	0.013	0.065	0.131	**0.509**	0.011	0.055	**0.111**	0.506	0.010	0.052	0.103
Mistral-Nemo	sentence-pair	0.510	0.014	0.069	0.073	0.504	**0.011**	**0.057**	0.066	0.501	0.010	0.051	0.101	0.502	0.010	0.051	0.101	0.518	**0.012**	0.059	0.118
	single-sentence	0.489	0.009	0.044	0.089	0.470	0.008	0.040	0.081	0.494	0.008	0.038	0.040	0.493	0.009	0.046	0.091	0.498	0.009	0.046	0.049
WizardLM-2 -8x22B	sentence-pair	0.558	0.012	0.059	0.119	**0.520**	**0.011**	0.054	**0.108**	0.508	0.010	0.052	0.103	0.510	0.010	0.052	0.104	**0.551**	**0.012**	**0.061**	**0.122**
	single-sentence	0.501	0.010	0.050	0.101	0.505	0.011	0.053	0.105	0.505	0.013	0.045	0.045	0.504	**0.012**	0.049	0.049	0.499	0.009	0.047	0.048

Note: AUC = AUC-ROC, T1 = TPR@1%FPR, T5 = TPR@5%FPR, T10 = TPR@10%FPR. Single-sentence evaluation uses 50–50 sampling. Best (**red bold**) and second-best (blue underlined) results are marked within each setup for each task and metric.

realistic use cases (Appendix D): (1) **Single-Sentence Exhaustive**: all samples with balanced 50-50 distribution;
(2) **Sampling (30-70)**: 30% positive, 70% negative;
(3) **Sampling (50-50)**: balanced random sampling;
(4) **Sampling (80-20)**: 80% positive, 20% negative;
(5) **Sentence-Pair Recognition**: pairwise comparison with random order presentation.
Full implementation details are provided in Appendix D.

5 Results

Tables 4 and 5 report complete results across 4 zero-shot and 7 model-based detectors across five tasks and evaluation setups.

Task 1 (Paraphrase Source Attribution). Both detector categories struggle with absolute classification (AUC 0.46–0.52 single-sentence). RADAR achieves the best results (AUC 0.728 sentence-pair, 0.648 exhaustive), followed by Kimi-K2-Instruct (AUC 0.691 sentence-pair). The improved sentence-pair performance suggests that comparative signals partially compensate for the difficulty of distinguishing human paraphrases (Type 3) from LLM paraphrases (Type 4) in isolation.

Task 2 (General Authorship Detection). RADAR dominates with AUC 0.910 (sentence-pair) and 0.797 (exhaustive). Model-based detectors underperform substantially (best: Kimi AUC 0.540 single-sentence), suggesting instruction-following LLMs cannot exploit representation-space differences between $\mathcal{X}_{\text{human}}$ and $\mathcal{X}_{\text{gen}}$ without task-specific fine-tuning.

Task 3 (AI Text Laundering Detection). Performance degrades substantially across most detectors. RADAR maintains moderate sentence-pair capability (AUC 0.748), but this represents a 16-point drop from its Task 2 performance—consistent with the authorship obfuscation blind spot identified in Section 2. Single-sentence performance collapses to near-random across all detectors (AUC

0.47–0.54), confirming that once texts undergo iterative paraphrasing, source origin becomes unreliable for absolute classification.

Task 4 (Iterative Depth Detection). Near-random performance across all detectors (AUC 0.487–0.529). Rather than a detector failure per se, this is a fundamental consequence of the small semantic difference between 1-iteration and 3-iteration paraphrasing identified in Section 2. Task 4 serves as a deliberate stress test probing the detection floor, motivating future depth-aware detection methods.

Task 5 (Paraphrase Attack Detection). RADAR demonstrates strong performance (AUC 0.909 sentence-pair, 0.803 exhaustive), confirming that deeply laundered AI text preserves detectable generation artifacts against $\mathcal{X}_{\text{human}}$ baselines. Model-based detectors show modest capability (best: Kimi AUC 0.573 single-sentence) but cannot match RADAR's fine-tuned sensitivity to generation-level patterns.

Common Observations.

Two Attack Categories Empirically Confirmed. Task 3's difficulty (best AUC 0.748; near-random for most detectors) versus Task 5's tractability (best AUC 0.909) empirically validates the asymmetry predicted by the mechanistic analysis: both tasks involve iteratively paraphrased text, but authorship obfuscation (Task 3) erases the source signal while plagiarism evasion (Task 5) preserves detectable generation artifacts. The gap between these two tasks is the central empirical finding of PADBen.

Representation Space Asymmetry. RADAR's consistent superiority across Tasks 2, 3, and 5 reflects its adversarial training on generation-level patterns that instruction-following models cannot access through prompting alone—confirming that generation-level features require task-specific fine-tuning to exploit.

Evaluation Robustness. Single-sentence sampling variants show stable performance (variation ≤ 0.02 AUC across all splits), confirming results reflect stylistic discrimination rather than base-rate exploitation. Zero-shot detectors consistently benefit from sentence-pair evaluation (+0.1–0.2 AUC), while model-based detectors show inconsistent patterns attributable to nondeterministic generation.

6 Conclusion

This paper presents PADBen, the first benchmark to systematically evaluate AI text detectors against iterative paraphrase attacks across two distinct scenarios: authorship obfuscation and plagiarism evasion. Evaluation of 11 detectors across five tasks confirms the predicted asymmetry: plagiarism evasion remains detectable (RADAR AUC 0.909) while authorship obfuscation substantially degrades all detectors (best AUC 0.748, near-random for most). These findings demonstrate that current binary human-vs-AIGT classifiers still lack robustness when facing the paraphrase attack setting, motivating detection methods that go beyond binary semantic and stylistic discrimination. The mechanism analysis relies on BGE-M3 embeddings and Qwen3-4B hidden states; quantitative replication with other models and direct measurement of lexical/syntactic feature stability are reserved for future work. Extending PADBen to deeper iteration depths and additional paraphrasing models would further characterize the detection floor identified in Task 4.

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